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Case 36-2022: A 30-Year-Old Woman with Decreased Vision and Headache

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PRESENTATION OF CASE

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Dr. Bart K. Chwalisz: A 30-year-old woman was evaluated in the neuro-ophthalmology clinic at a hospital affiliated with this hospital because of monocular vision loss.

Two months before the current presentation, during the third trimester of the patient's first pregnancy, headache and decreased vision in the right eye developed. The patient thought that the headache felt like previous migraines, which had been infrequent. However, decreased vision had not been a typical feature of her migraines. Colors appeared dim, and light perception was decreased. There was soreness in the right eye but no pain with eye movement.

Five days later, the patient was evaluated by an optometrist. She reported that she could see only shapes and movements. On a visual field test, there was a normal response in the left eye but no response in the right eye. The patient was instructed to seek evaluation in an emergency department of another hospital.

In the emergency department, magnetic resonance imaging (MRI) of the head, which was performed without the administration of intravenous contrast material because of the pregnancy, revealed marked enlargement of the intraorbital right optic nerve. The nerve appeared isointense as compared with the extraocular muscles on the T1-weighted sequences but appeared bright on the T2-weighted sequence; the globes appeared normal, and the extraocular muscles were not enlarged. The results of magnetic resonance venography of the head were normal.

The patient was admitted to the hospital, and treatment with intravenous methylprednisolone was initiated for a working diagnosis of optic neuritis. During the hospitalization, her ability to see colors improved slightly, but her vision did not improve; she could see only hand motions. On the fifth hospital day, she was discharged home with instructions to schedule follow-up appointments with a neurologist and an optometrist.

One month after discharge from the other hospital and 2 weeks before the current presentation, the patient was again evaluated by her optometrist. The vision in the right eye had worsened; she was no longer able to see shadows or movements. The patient was referred for evaluation by a neuro-ophthalmologist.

On the current presentation to the neuro-ophthalmology clinic of a hospital affiliated with this hospital, the patient reported no improvement in vision and ongoing soreness in the right eye. There was a history of Lyme disease, which had been diagnosed during the second trimester of pregnancy and had been treated with amoxicillin. Medications included prenatal vitamins and ferrous sulfate. The patient lived with her husband in New England and worked in health care. She had never smoked tobacco and did not drink alcohol or use illicit drugs. A brother had a history of childhood epilepsy, and her mother had macular degeneration.

On examination, the visual acuity was 20/15 in the left eye, and there was no light perception in the right eye. The pupils were symmetric and reactive to light, although the response in the right pupil was sluggish; an afferent pupillary defect was present in the right eye. On the Ishihara color test, all plates were identified with the left eye. On a visual field test, there was a normal response in the left eye. The Ishihara color test and the visual field test could not be performed on the right eye because of poor vision. The results of a slit-lamp examination were normal, with no cells identified in the anterior chamber or vitreous. On a funduscopy examination (Fig. 1A and 1B), the optic disk in the right eye was swollen and had temporal pallor; there was a small area of hypopigmentation close to the macula in the left eye. Optical coherence tomography revealed increased thickness of the retinal nerve fiber layer and circumferential thinning of the ganglion cell complex of the right eye; no abnormalities were noted in the left eye.

The blood creatinine level was 0.56 mg per deciliter (50 μ mol per liter; reference range, 0.60 to 1.50 mg per deciliter [53 to 133 μ mol per liter]) and the blood alkaline phosphatase level 147 U per liter (reference range, 30 to 100). The results of other kidney-function and liver-function tests were normal, as were the blood levels of electrolytes, the complete blood count, and the results of coagulation tests. The blood C-reactive protein level was 15 mg per liter (reference value, <8) and the erythrocyte sedimentation rate 34 mm per hour (reference range, 0 to 20). A blood test for antinuclear antibodies was positive at a titer of 1:40 with a speckled pattern. Tests for antibodies against aquaporin-4 and myelin oligodendrocyte glycoprotein (MOG) were negative. A

diagnosis of atypical optic neuritis was considered. Because the patient was near delivery, further treatment was delayed until after the birth of her child, which occurred 3 days later.

Six days after delivery of the baby, the patient was admitted to this hospital, and additional studies were obtained. Lumbar puncture was performed. On cerebrospinal fluid (CSF) analysis, there were no nucleated cells, and the levels of protein, glucose, albumin, and IgG were normal. There was no oligoclonal banding after concentration of the CSF. Testing of the CSF was negative for syphilis antibodies and for antibodies against aquaporin-4.

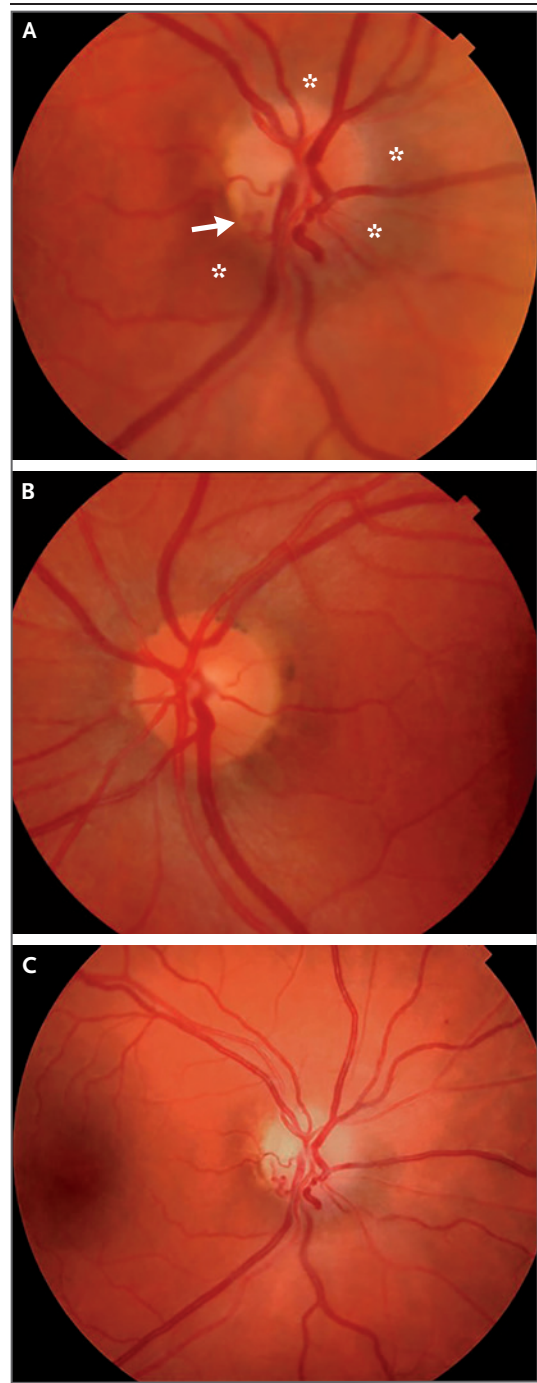
Dr. Benjamin M. Kozak: MRI of the head (Fig. 2), performed before and after the administration of intravenous contrast material, revealed prominent fusiform enlargement of a segment of the intraorbital right optic nerve with associated enhancement, as well as associated hyperintensity on the T2-weighted short-tau inversion recovery (STIR) sequence. There was no evidence of associated orbital edema or other orbital abnormalities. There was a punctate hyperintense focus within the left parietal subcortical white matter on the T2-weighted fluid-attenuated inversion recovery sequence, without associated enhancement or restricted diffusion.

Dr. Chwalisz: Treatment with intravenous methylprednisolone and daily plasmapheresis was administered. On the fourth hospital day, the soreness in the right eye had resolved, and there was light perception in the right eye. On the fifth hospital day, treatment with intravenous methylprednisolone was changed to oral prednisone. On the ninth hospital day, rituximab was administered, and the patient was discharged home with a prescription for a 7-week tapering course of prednisone.

During the next 3 weeks, treatment with rituximab was administered weekly. Two weeks after the last rituximab infusion, the patient was evaluated in the neuro-ophthalmology clinic at the hospital affiliated with this hospital. She reported no eye pain and only occasional headache, but she noted that the vision in the right eye had worsened. Examination revealed no light perception in the right eye, with an afferent pupillary defect. The results of a visual field test and the Ishihara color test were unchanged from the previous evaluation in the neuro-ophthalmology clinic. On a funduscopy examination

Figure 1. Fundusoscopic Images.

A fundusoscopic examination was performed 8 weeks after the onset of symptoms. On a photograph of the right eye (Panel A), the optic disk shows pallor as compared with the optic disk of the left eye (Panel B), as well as blurred margins due to swelling (asterisks) and subtle evidence of abnormal vessels in the nasal aspect (arrow). A follow-up fundusoscopic examination was performed 16 weeks after the onset of symptoms. On a photograph of the right eye (Panel C), the optic disk shows increased pallor, decreased swelling, and greater prominence of the abnormal vessels in the nasal aspect, a finding consistent with opticociliary shunt vessels.



(Fig. 1C), the optic disk in the right eye was swollen and had more pronounced temporal pallor; opticociliary shunt vessels were also newly present. Additional imaging studies were obtained.

Dr. Kozak: Follow-up MRI of the head (Fig. 2) revealed no radiologically significant change in the segmental enlargement of the intraorbital right optic nerve, with persistent enhancement as well as hyperintensity on the T2-weighted STIR sequence. No new orbital abnormalities were present. In addition, the punctate lesion in the left parietal subcortical white matter was unchanged.

Dr. Chwalisz: A diagnostic procedure was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Michael Levy: This young woman presented with subacute vision loss. The first step in the evaluation of this patient is to localize the problem to the eye, the optic nerve, or the brain. The patient was first evaluated by an eye specialist, who examined the eye for conditions that can lead to monocular vision loss.¹ When examination of the eye did not identify a cause of her vision loss, she was next evaluated for processes that affect the optic nerve or brain. This evaluation needed to begin quickly to avoid further damage from a potentially treatable disease.

The optic nerve fibers extend 50 mm caudally from the retina to the optic chiasm. A lesion anywhere along the optic nerve can cause monocular vision loss. In the optic chiasm, which is wedged between the hypothalamus of the brain above and the pituitary gland below, the optic tracts begin to segment into the right and left visual fields. Vision loss in either the right or the

left visual field localizes to a lesion in the optic chiasm, the optic tracts, the optic radiations in the brain, or the visual cortex. In this patient, the examination revealed monocular vision loss — and not a visual field defect in both eyes. I will focus the differential diagnosis on processes that affect the optic nerve.

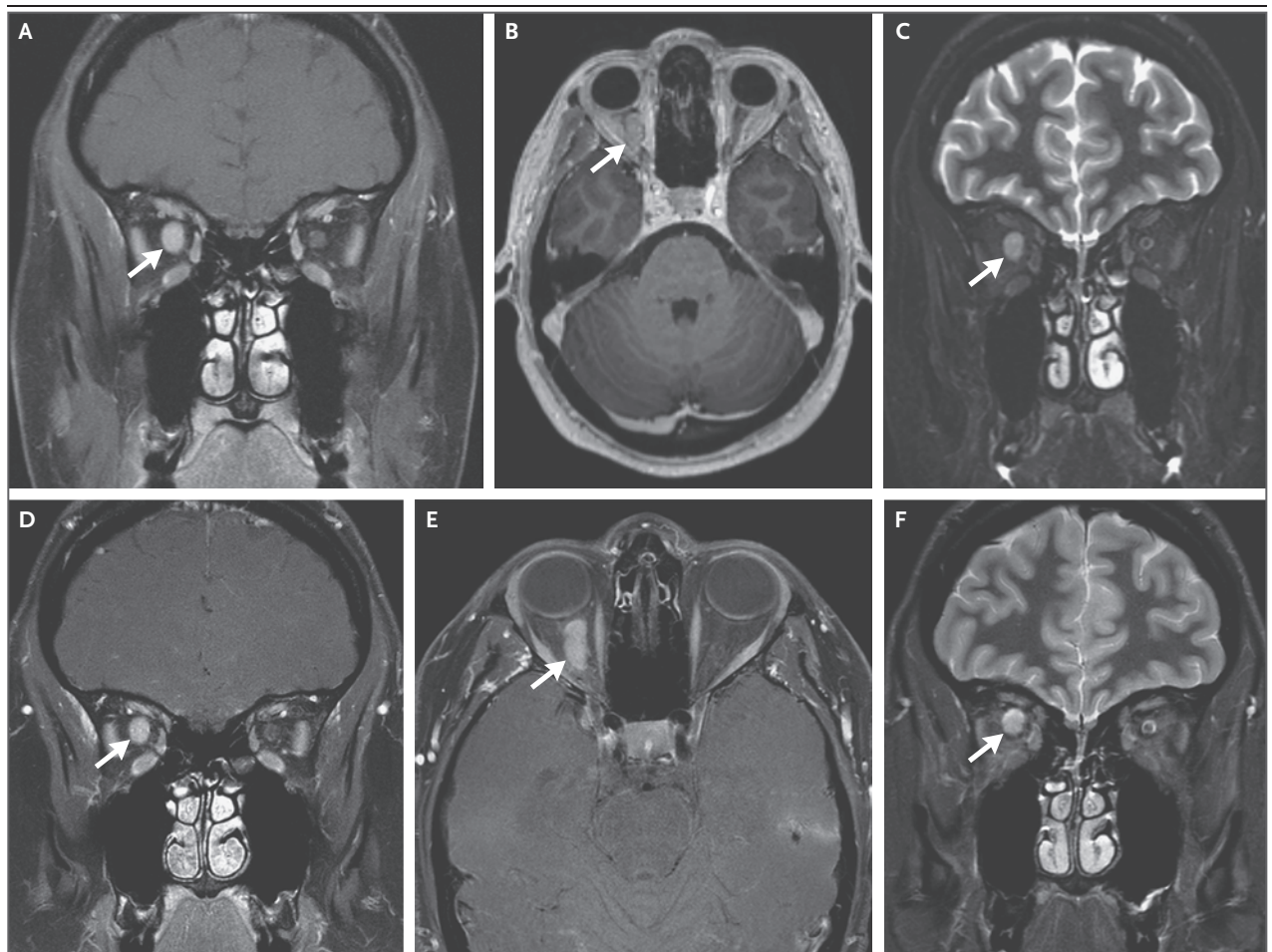


Figure 2. MRI of the Head.

On admission to this hospital, approximately 9 weeks after the onset of symptoms, MRI of the head was performed before and after the administration of intravenous contrast material. Coronal and axial contrast-enhanced T1-weighted sequences (Panels A and B, respectively) show prominent fusiform enlargement of a segment of the intraorbital right optic nerve with abnormal increased homogeneous enhancement (arrows). A coronal T2-weighted short-tau inversion recovery (STIR) sequence (Panel C) shows diffuse hyperintensity involving this segment of the right optic nerve (arrow) without adjacent orbital edema. Approximately 2 months later, follow-up MRI of the head was performed before and after the administration of intravenous contrast material. Coronal and axial contrast-enhanced T1-weighted sequences (Panels D and E, respectively) show similar segmental enlargement and abnormal enhancement of the intraorbital right optic nerve (arrows). A coronal T2-weighted STIR sequence (Panel F) shows similar hyperintensity involving this segment of the right optic nerve (arrow) without inflammatory changes in the orbital fat.

OPTIC NEURITIS

In a young woman, the differential diagnosis for diseases that affect the optic nerve starts with inflammatory disorders. When considering possible causes of inflammatory optic neuritis, it is useful to divide the optic nerve into three segments: anterior, intracanalicular, and posterior.

The anterior segment of the optic nerve is 25 mm in length and extends from the retina to the entrance of the skull. This segment is commonly the target of multiple sclerosis.² A total of

56% of patients with multiple sclerosis have involvement of the optic nerve at some point in the disease course.³ The lesions are usually focal (i.e., not long) and located within the optic nerve itself. Optic neuritis is the initial event in 20% of patients who present with a new diagnosis of multiple sclerosis; for this reason, the workup for optic neuritis includes MRI of the head to look for the periventricular lesions that are typical of multiple sclerosis.⁴ In patients with multiple sclerosis, optic neuritis usually heals with-

out treatment, but long-term therapy to prevent additional inflammatory attacks in the central nervous system reduces the risk of long-term disability.⁵ In this patient, MRI of the head did not show lesions that would be consistent with a diagnosis of multiple sclerosis. In addition, she did not have features associated with a high risk of this diagnosis, such as a family history; if such features had been present, yearly MRI of the head may have been indicated. My overall suspicion for multiple sclerosis in this patient is low.

The anterior segment of the optic nerve is also a target of monophasic optic neuritis (also called idiopathic optic neuritis). Episodes often occur after an immunologic response is triggered by infection or the administration of a vaccine. Monophasic optic neuritis is usually associated with a high likelihood of recovery and a low risk of relapse. There is no diagnostic test for monophasic optic neuritis, and the diagnosis is typically considered when alternative causes are ruled out. Multiple sclerosis and monophasic optic neuritis together account for 95% of cases of optic neuritis; each of these diagnoses is equally likely to be the cause of optic neuritis.⁴ Less common causes of optic neuritis include neurosarcoidosis, MOG antibody disease (MOGAD), and neuromyelitis optica spectrum disorder (NMOSD).

Neurosarcoidosis is a systemic granulomatous process, whereas MOGAD and NMOSD are antigen-specific autoimmune diseases that target MOG and the water channel aquaporin-4, respectively. Patients with neurosarcoidosis can present with a subacute course of optic neuritis, which often develops in the context of other cranial neuropathies, because the inflammation tends to spread along the basal meninges.⁶ In patients with MOGAD, lesions involving the optic nerve tend to be long, sometimes extending across multiple segments of the optic nerve, and they cause meningeal inflammation (i.e., perineuritis) that leads to concomitant pain.⁷ In patients with NMOSD, the intracanalicular and posterior segments of the optic nerve are common targets of inflammation. The intracanalicular segment of the optic nerve is only 10 mm long, but inflammation in this part of the nerve is problematic because of the limited space for swelling within the bony canal. NMOSD lesions are among the most destructive lesions associated with inflammatory diseases: optic neuritis due to NMOSD causes blindness in 60% of pa-

tients despite treatment with immunosuppressive therapy.⁸

Both MOGAD and NMOSD are considerations in this patient because of the severe vision loss and the extensive swelling and enhancement of the optic nerve on MRI. For each of these diagnoses, serologic tests are highly specific in the context of optic neuritis; the sensitivity of these assays ranges from 60 to 94%.^{9,10} In this patient, the blood tests for antibodies against MOG and aquaporin-4 were negative. Although seronegative disease is possible, I will consider other potential explanations for this patient's vision loss.

INFECTION

Infectious optic neuropathies are not common in the United States. The two most likely causes in this patient are *Bartonella henselae* and *Toxoplasma gondii*. When these organisms affect the optic nerve, inflammation of the retina typically also occurs, and this finding is readily identified on funduscopy. In this patient, funduscopy showed signs of venous congestion of the optic nerve but no evidence of infection.¹¹

TUMOR

MRI in this patient revealed a lesion in the optic nerve that had a masslike appearance and did not decrease in size despite the administration of intravenous methylprednisolone. The presence of persistent or treatment-refractory optic neuritis is suggestive of NMOSD or sarcoid optic neuropathy. However, the presence of a persistent mass lesion of the optic nerve can be indicative of a tumor, most commonly an optic nerve glioma or an optic nerve sheath meningioma.

Optic nerve gliomas arise from astrocytes and form low-grade masses similar to pilocytic astrocytomas of the inferior fossa.¹² More than 90% of optic nerve gliomas occur in persons younger than 20 years of age.¹³ In this case, the lesion seen on MRI had the classic appearance of an optic nerve glioma, but the patient was 30 years of age.

Optic nerve sheath meningiomas arise from the meningeal tissue around the optic nerve fibers and typically compromise vision by compressing the optic nerve over time.¹⁴ Because optic nerve sheath meningiomas do not typically affect optic nerve tissue, MRI may show a characteristic "tram-track" sign, a pattern caused by

simultaneous enhancement of tumor tissue and nonenhancement of the optic nerve.¹⁵ Although the tram-track sign was not observed on MRI in this patient, optic nerve sheath meningiomas are more likely to occur in adults than optic nerve gliomas. For both of these tumors, the diagnosis can be made by means of biopsy, and treatment involves surgical resection or radiation therapy. Surgery is associated with a high risk of blindness in the affected eye.

Biopsy of the optic nerve sheath is rarely performed in the workup for optic neuropathies because of the risk of permanent vision damage. However, in this patient, the poor response to immunosuppressive therapy and the persistent enhancement and masslike appearance of the lesion on MRI make a biopsy necessary for diagnosis. I suspect that the diagnostic test in this patient was a biopsy of the right optic nerve sheath.

CLINICAL IMPRESSION

Dr. Chwalisz: This young, pregnant woman presented with subacute painful monocular vision loss. Early dyschromatopsia progressed to severely decreased visual acuity with a relative afferent pupillary defect, features suggestive of an optic neuropathy. This diagnosis was supported by an abnormal appearance of the optic disk, which had swelling, pallor, and optociliary shunt vessels. An atypical optic neuritis was the initial presumptive diagnosis, but the patient's vision did not change after the administration of immunotherapy. On repeat MRI, the optic nerve was too enlarged and the enhancement too prolonged for a diagnosis of optic neuritis. The normal results of serologic testing and CSF analysis further decreased the probability of inflammation, infection, or infiltration by hematologic cancer. Therefore, we considered an intrinsic optic nerve tumor to be the most likely diagnosis.

The identification of optociliary shunt vessels provided an important clue in this case. Optociliary shunt vessels form when drainage of the central retinal vein is occluded and blood flows through collateral veins to peripapillary chorioidal veins.¹⁶ Optociliary shunt vessels occur occasionally in the context of other conditions, but the classic and pertinent association is with optic nerve sheath meningiomas. Meningiomas

may express progesterone receptors and may grow at an accelerated pace during pregnancy.¹⁷ However, MRI did not show the typical radiologic appearance of an optic nerve sheath meningioma; in fact, the findings were more suggestive of a glioma.¹⁵ Thus, a biopsy of the optic nerve sheath was recommended to distinguish between these two diagnoses.

CLINICAL DIAGNOSIS

Optic nerve tumor, either optic nerve sheath meningioma or optic nerve glioma.

DR. MICHAEL LEVY'S DIAGNOSIS

Optic nerve glioma or optic nerve sheath meningioma.

DIAGNOSTIC PROCEDURE

Dr. Michael K. Yoon: A biopsy of the optic nerve sheath is rarely performed because in approximately 2% of patients, the procedure results in blindness due to vascular injury to the optic nerve. Because this patient had long-standing vision loss and optic nerve pallor, vision recovery was not expected, so vision loss as a complication of the biopsy was not considered to be a clinically significant risk.

There are several approaches to performing a biopsy of the optic nerve sheath, but entry through the medial upper eyelid is the most common. This approach follows the shortest route to the nerve and does not require detachment of any extraocular muscles or removal of bone. In addition, the incision is placed in the eyelid crease, which has the most cosmetically acceptable outcome after healing is complete.

In this patient, we performed the biopsy after general anesthesia was established. An operating microscope was used (Fig. 3). A 15-mm incision was made in the crease of the right medial upper eyelid. Once the orbital septum was incised, blunt dissection was performed until the nerve was visualized. The optic nerve sheath was approximately twice the diameter of a normal optic nerve sheath and had an irregular and highly vascular appearance. A small portion of the optic nerve sheath was sampled. Despite the patient's vision loss, a complete transection of the nerve was not performed.

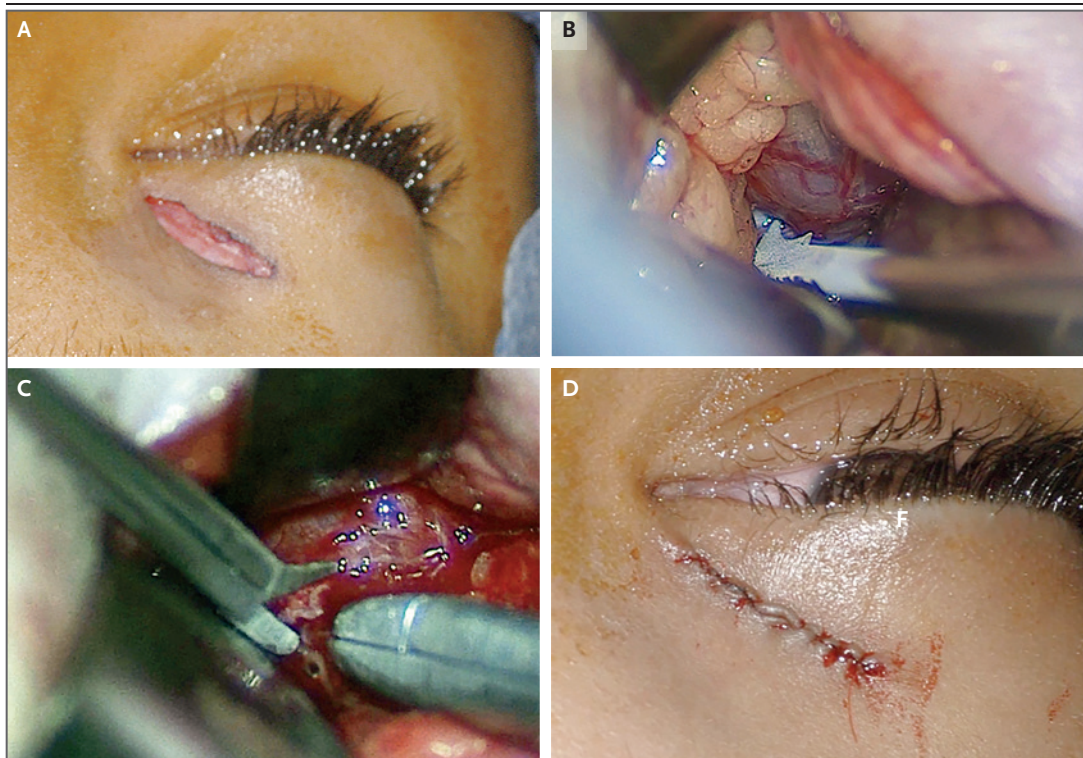


Figure 3. Photographs Obtained during Biopsy of the Optic Nerve Sheath.

Photographs were obtained through an operating microscope from the surgeon's view (inverted). Panel A shows the incision through the right medial upper eyelid. Panel B shows the optic nerve sheath, which is gray and twice the size of a normal optic nerve sheath. Panel C shows the thickened optic nerve sheath after incision, with hyper-vascularity that has led to bleeding. Panel D shows closure of the incision with gut sutures at the conclusion of the procedure.

PATHOLOGICAL DISCUSSION

Dr. Anna M. Stagner: Hematoxylin and eosin staining of sections of the orbital biopsy specimen (Fig. 4) showed fragments of an eosinophilic neoplasm admixed with orbital fibroadipose tissue and a small fragment of bone. The tumor had a distinctly lobulated or nested architecture. The neoplastic cells appeared to grow in a syncytial pattern and had indistinct cell membranes, abundant eosinophilic cytoplasm, and relatively uniform round-to-oval nuclei. Intracellular pseudoinclusions were identified, and concentric lamellated calcifications (psammoma bodies) were not detected. There were no features of a malignant tumor (World Health Organization [WHO] grade 3) or atypical meningioma (WHO grade 2), such as increased mitotic

activity, glial invasion, increased cellularity, sheetlike growth, prominent nucleoli, or necrosis; the absence of such features was consistent with a low-grade tumor (WHO grade 1). The tumor cells showed diffuse nuclear staining for the progesterone receptor, which is a common feature of low-grade meningiomas, particularly those involving the optic nerve sheath.^{18,19} The Ki-67 proliferation index — an important prognostic measure for determining the risk of recurrence, even in patients with otherwise low-grade tumors — was very low, at 0.7% (8 positive cells out of 1080 cells counted).²⁰

PATHOLOGICAL DIAGNOSIS

Meningioma, meningothelial type, World Health Organization grade 1.

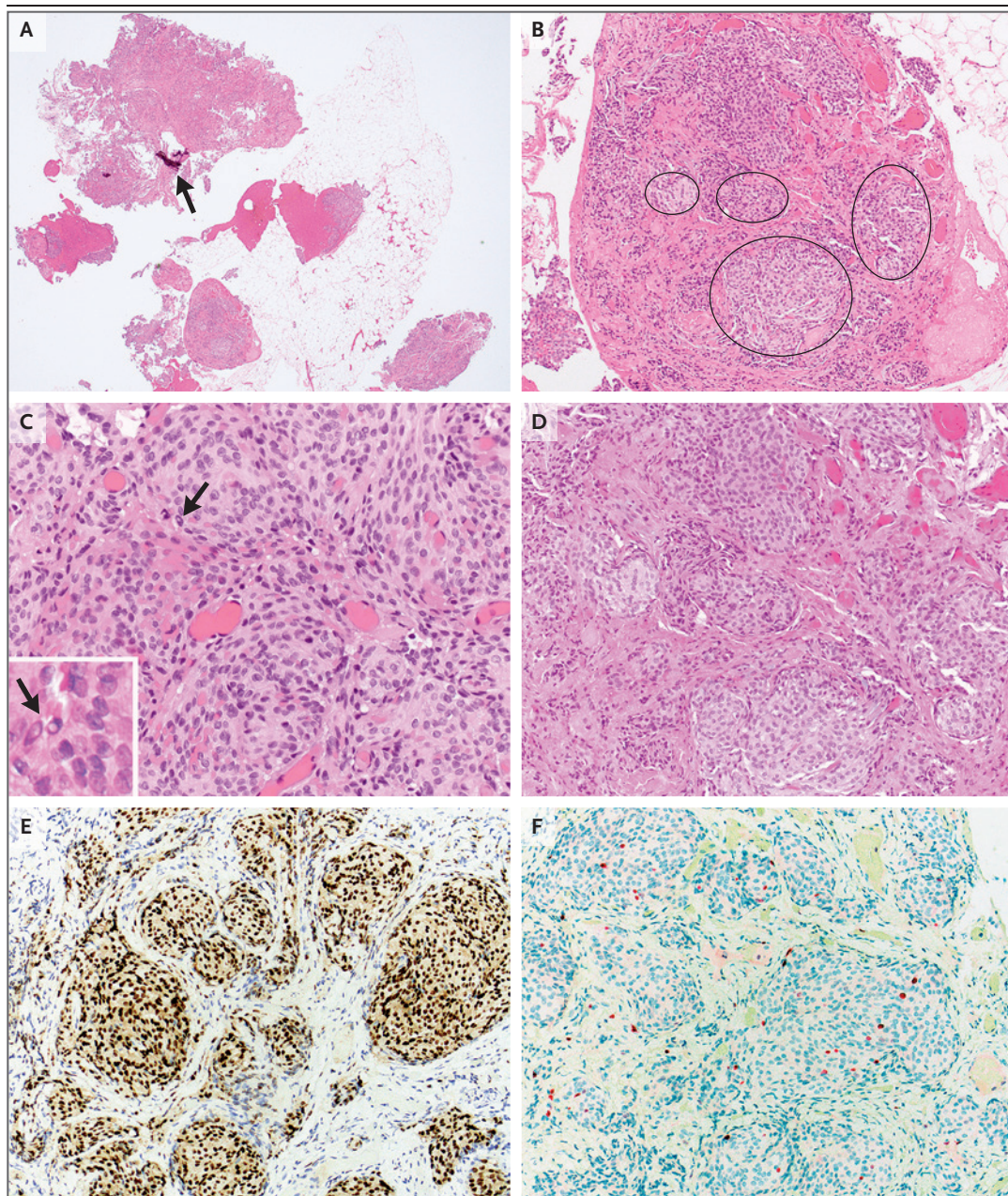


Figure 4. Biopsy Specimen of the Optic Nerve Sheath.

Hematoxylin and eosin staining shows fragments of an eosinophilic tumor admixed with orbital fat and bone (Panel A, arrow). At higher magnification, the tumor architecture is predominantly nested or whorled (Panel B, circled); this growth pattern is characteristic of meningothelial meningioma. The cells have indistinct cell borders and monotonous, banal, round-to-oval nuclei with occasional nuclear pseudoinclusions (Panel C and inset, arrows). There is no evidence of atypical features, such as necrosis, mitotic figures, or glial invasion (Panel D). On immunohistochemical staining for progesterone receptor, there is diffuse nuclear positivity (Panel E, in brown). On immunohistochemical staining for Ki-67, the proliferation index is very low, with positive cells constituting less than 1% of the cells counted (Panel F, in red).

DISCUSSION OF MANAGEMENT

Dr. Helen A. Shih: Optic nerve sheath meningiomas are typically managed with radiation therapy.²¹ The timing of radiation therapy for a patient with an optic nerve sheath meningioma is most commonly based on clinical and imaging findings, given that biopsy is not usually attempted because of the risk of further vision impairment. However, if a histologic diagnosis is available, it can be useful in affirming the clinical assessment and management plan.

The primary goal of radiation therapy is to prevent further tumor growth. Unchecked tumor growth leads to vision loss and gradual proptosis of the eye, which causes disfigurement and pain and eventually leads to enucleation. In most retrospective series, radiation therapy has resulted in durable tumor control in more than 90% of cases.^{21,22} Beyond tumor control and prevention of cosmetic changes, additional goals of treatment are vision retention and sometimes vision recovery, which are most likely to occur if treatment is initiated within weeks to months after the onset of symptoms.²² Patients with long-standing vision impairment do not typically have vision recovery.

In this patient, radiation therapy was administered 5 days per week over a 6-week period, with a total dose delivered of 54 Gy in 30 fractions. To minimize the long-term side effects associated with radiation therapy, collateral irradiation of the anterior globe, lacrimal gland, surrounding normal brain tissue, contralateral orbit, optic chiasm, and pituitary gland was limited. Proton therapy was used to deliver the lowest level of collateral radiation to the surrounding normal tissues. The patient had a reasonably good response to treatment, although she had expected side effects, including local pain and pressure, ptosis, lagging eye movements, and fatigue. One year after radiation

therapy, tumor control was confirmed on imaging and her symptoms were much less severe, but she had not regained any vision.

PATIENT'S PERSPECTIVE

Patient: I was in the third trimester of my pregnancy and had this throbbing headache while driving. When I would rub my left eye and use my right eye to see, all the headlights on the cars seemed to turn off. By the time the tumor was diagnosed, several months had passed. I never thought I would fall through the cracks, but I almost did. Going through treatment was overwhelming. I was told initially that it wasn't a tumor and I would get my vision back. All at once, I had the experience of being a new mom, undergoing treatment, maintaining hope for vision recovery, managing the effect on my career, and spending Christmas hospitalized with a newborn — I was in a fog during that period. It took a year to reflect on my experience. I regret not pushing harder to be transferred to a larger center when I was first hospitalized, and I wonder whether my vision might have recovered if I had. When I was in treatment, my providers made my postpartum goals a priority. It relieved the guilt of not being present during maternity leave. If you take anything away from my experience, I hope it's to treat vision loss with urgency because it could mean vision recovery. This diagnosis changed my life, but I'm lucky. Thank you to my team. I feel lucky to have you by my side.

FINAL DIAGNOSIS

Meningioma, meningothelial type, World Health Organization grade 1.

This case was presented at Neurology Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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